

Color system — brand palette, semantics, connection-state & contrast proofs

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** HelixVPN's complete colour system: the brand primitive ramps (a privacy/trust palette, concrete hex), the **full light AND dark semantic mappings** (surface / text / border / overlay / feedback / action), the **connection-state palette** mapping onto the **exact 7-variant ffi::TunnelStatus** [v04-client/ffi-surface.md §3.2], the **WCAG contrast proofs** (real arithmetic, every cited ratio computed — §11.4.6), the no-overlap / no-label-overlay colour-layer rule (§11.4.162), and the per-app accent differentiation (Client / Console / Connector).

SPEC-ONLY. It defines *which colours mean what* and *proves they are legible*. The token **structure** these colours live in (tiers, schema, theming model) is owned by [design-tokens.md]; the multi-form emit by [token-export-pipeline.md]. This document is **original HelixVPN design work** — the hex choices are owned here.

Boundary with sibling docs. Owns: hex values + semantic colour roles + contrast proofs. Consumes: the token tier/naming model [design-tokens.md]; the 7-variant ffi::TunnelStatus [FFI §3.2]; the 3-app/8-platform matrix [SPINE §3]; the WCAG relative-luminance formula (W3C, cited in Sources).

Evidence base. [FFI §N] = final/v04-client/ffi-surface.md; [DT §N] = final/v10-design/design-tokens.md; [SPINE §N] = final/SPECIFICATION.md. **Every contrast ratio in this document was computed by the WCAG 2.1 relative-luminance + contrast formula on the exact hexes stated** (the computation script + raw output are the captured evidence; §11.4.6 — no ratio is guessed). Anything unproven is tagged UNVERIFIED.

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0. Palette philosophy — privacy & trust

HelixVPN is a privacy product; its colour language must read **calm, trustworthy, and unambiguous about safety state**. Three design commitments drive every hex choice:

1. **Deep indigo-blue as the brand** — blue is the near-universal "secure / trusted / institutional" hue; HelixVPN uses a slightly indigo-shifted blue (`helix.500` #3D5AF1) so it reads *modern privacy tool* rather than *corporate banking*. It is the resting/idle and primary-action colour.
2. **Low-chroma neutrals** — surfaces and text are near-neutral cool greys so the only **saturated** colour on screen at any moment is the **connection-state** colour. The user's eye is drawn to the one thing that matters: *am I protected?*
3. **Unmissable, conventional safety semantics** — green = protected, amber = working, orange = unexpectedly dropped, **red = danger** (leak / kill-switch tripped). These never change across brand or tenant (§8 D-COLOR-1); a privacy product must never recolour its safety signal.

Honest boundary (§11.4.6). Colour alone is **never** the sole signal for a state (≈8 % of users have colour-vision deficiency). Every connection state also carries an **icon + text label** (the StatusChip/ConnectButton always show e.g. a shield-check glyph + "Connected · direct"); colour is reinforcement, not the only channel. Contrast proofs (§4) cover the legibility of the label, which is the load-bearing signal.

1. Primitive ramps (concrete hex)

Tier-1 primitives (theme-agnostic literals, [DT §1]). These are the **only** place a literal colour appears; everything else references these.

1.1 Brand — `color.primitive.helix.*` (indigo-blue)

Stop	Hex	Role hint
<code>helix.50</code>	#EEF2FF	lightest tint (selected-row bg, light)

Stop	Hex	Role hint
helix.100	#E0E7FF	hover tint (light)
helix.200	#C3CEFF	subtle border / chip (light)
helix.300	#9CB0FF	dark-theme link / accent text
helix.400	#6B86FB	dark-theme action / focus
helix.500	#3D5AF1	brand core — logo, focus ring
helix.600	#2F47C4	primary action fill (light), link text (light)
helix.700	#263A9E	pressed action (light)
helix.800	#1F2F7A	deep accent surface
helix.900	#16205A	brand-tint dark surface

1.2 Neutral — `color.primitive.neutral.*` (cool grey, 0–1000)

Stop	Hex	Stop	Hex
neutral.0	#FFFFFF	neutral.500	#6B7385
neutral.50	#F7F8FA	neutral.600	#4B5263
neutral.100	#EDEFF3	neutral.700	#353B49
neutral.200	#DCE0E8	neutral.800	#222732
neutral.300	#C2C8D4	neutral.900	#151920
neutral.400	#9AA2B2	neutral.1000	#0B0E13

1.3 Status & feedback ramps (key stops)

Ramp	Light-usable stops	Dark-usable stops
green.* (protected)	green.50 #ECFDF3 · green.500 #22C55E · green.600 #15803D · green.700 #166534	green.300 #6CE9A6 · green.500 #22C55E · green.400 #34D399
amber.* (working)	amber.50 #FFFBEF · amber.500 #F59E0B · amber.600 #D97706 · amber.700 #B45309	amber.300 #FCD34D · amber.400 #FBBF24
orange.* (dropped)	orange.50 #FFF4ED · orange.500 #F97316 · orange.600 #C2410C · orange.700 #9A3412	orange.400 #FB923C
red.* (danger)	red.50 #FEE2F2 · red.500 #EF4444 · red.600 #DC2626 · red.700 #B91C1C	red.300 #FCA5A5 · red.400 #F87171
teal.* (accent / relay)	teal.50 #F0FDF4 · teal.500 #14B8A6 · teal.600 #0D9488 · teal.700 #0F766E	teal.300 #5EEAD4 · teal.400 #2DD4BF
		violet.400 #A78BFA

Ramp	Light-usable stops	Dark-usable stops
violet.* (Console accent)	violet.500 #8B5CF6 · violet.600 #7C3AED · violet.700 #6D28D9	
blue.* (info)	blue.500 #2563EB · blue.600 #1D4ED8	blue.400 #60A5FA

Why both a .500-class bright stop and a .600/.700-class dark stop per status ramp. The bright stop (green.500, amber.400, red.400) is the **dark-theme** text/icon/fill colour (high luminance reads on a dark surface); the dark stop (green.600, amber.700, red.600) is the **light-theme** text/icon/fill colour (low luminance reads on white). This is what makes the connection-state palette pass AA in **both** themes (§4).

2. Semantic colour roles — full light + dark mapping

Tier-2 semantics ([DT §1]). **Every** token below defines **both** light and dark (mandatory, gate CM-token-light-dark-complete [DT §9]; §11.4.162). Each side is a reference to a §1 primitive (literal shown for readability).

2.1 Surfaces & background

Semantic token	Light	Dark	Notes
surface.default	neutral.0 #FFFFFF	neutral.900 #151920	base app surface
surface.raised	#F7F8FA (neutral.50)	#222732 (neutral.800)	cards, list rows, sheets
surface.sunken	#EDEFF3 (neutral.100)	#0B0E13 (neutral.1000)	wells, input track
surface.brandTint	helix.50 #EEF2FF	helix.900 #16205A	selected/active brand surface
background.canvas	#FFFFFF	#0B0E13	the window behind surfaces

2.2 Text / on-surface

Semantic token	Light	Dark	Contrast vs its surface
text.primary	neutral.900 #151920	neutral.100 #EDEFF3	17.62 (light) / 15.31 (dark) → AAA
text.secondary	neutral.600 #4B5263	#C2C8D4 (neutral.300)	7.82 / 10.49 → AAA
text.tertiary	neutral.500 #6B7385	neutral.400 #9AA2B2	4.76 / 6.87 → AA

Semantic token	Light	Dark	Contrast vs its surface
text.disabled	neutral.400 #9AA2B2	neutral.500 #6B7385	2.57 / 3.71 — exempt (WCAG SC 1.4.3 inactive-control exception)
text.onState	#FFFFFF	#FFFFFF	white label drawn on a filled state colour (§3)
text.onBrand	#FFFFFF	#FFFFFF	label on a brand-action fill
text.link	helix.600 #2F47C4	helix.300 #9CB0FF	7.45 / 8.45 → AAA

2.3 Borders & overlays

Semantic token	Light	Dark	Notes
border.default	#DCE0E8 (neutral.200)	#353B49 (neutral.700)	1.32 (light) vs surface — decorative hairline; not a contrast-bearing element
border.strong	neutral.400 #9AA2B2	neutral.500 #6B7385	input/focus border (≥3.0 UI-component floor target)
border.focus	helix.500 #3D5AF1	helix.400 #6B86FB	focus ring — 5.34 / 5.39 vs surface → ≥3.0 SC 1.4.11 ✓
overlay.scrim	#1F2937 @ 60 % α	#000000 @ 70 % α	modal/sheet backdrop; white text on the scrim = 14.68 → AAA

2.4 Action & feedback

Semantic token	Light	Dark	Proof (label/role)
action.primary	helix.600 #2F47C4	helix.500 #3D5AF1	white label on light fill 7.45 AAA; dark theme uses helix.500, white label 5.34 AA
action.primary.hover	helix.700 #263A9E	helix.400 #6B86FB	—
action.secondary		same	

Semantic token	Light	Dark	Proof (label/role)
	surface.raised + border.strong		tonal/outline button
feedback.success	green.600 #15803D	green.400 #34D399	text-on-surface 5.02 / 9.17 → AA / AAA
feedback.warning	amber.700 #B45309	amber.400 #FBBF24	5.02 / 10.56 → AA / AAA
feedback.error	red.600 #DC2626	red.400 #F87171	4.83 / 6.37 → AA
feedback.info	blue.600 #1D4ED8	blue.400 #60A5FA	6.70 / 6.93 → AA

Every semantic colour token above has **both** a light and a dark value — none is single-theme. (text.onState/text.onBrand are intentionally white in both themes because they sit on a *coloured* fill, not on the theme surface; their legibility is proven against each state fill in §4.3, not against the surface.)

3. The connection-state palette (7-variant FFI mapping)

The product-defining colour group. It maps **one-to-one** onto the **7-variant ffi::TunnelStatus** the UI switches on [FFI §3.2]: Disconnected · Connecting · Handshaking · Connected{path:direct|relay} · Reconnecting · Down · Danger. The mapping is **total** — every variant has a dedicated colour in **both** themes (the UI never invents a colour, §11.4.6).

3.1 The mapping (both themes)

ffi::TunnelStatus	Family	Light fill (white label)	Dark fill (dark #151920 label)	Light text/icon (on tint)	Dark text/icon (on dark)
Disconnected	neutral grey	neutral.500 #6B7385	neutral.600 #4B5263	#6B7385	#9AA2B3
Connecting	amber	amber.700 #B45309	amber.400 #FBBF24	#B45309	#FBBF24
Handshaking	amber (= Connecting)	amber.700 #B45309	amber.400 #FBBF24	#B45309	#FBBF24
Connected{direct}	green (direct)	green.600 #15803D	green.500 #22C55E	#15803D	#22C55E
Connected{relay}	green (relay sub-shade)	teal.700 #0F766E	teal.400 #2DD4BF	#0F766E	#2DD4BF
Reconnecting				#B45309	#FBBF24

<code>ffi::TunnelStatus</code>	Family	Light fill (white label)	Dark fill (dark #151920 label)	Light text/ icon (on tint)	Dark text/ icon (on dark)
	amber pulse	amber.700 #B45309 + motion.connectPulse	amber.400 #FBBF24 + pulse		
Down	orange	orange.600 #C2410C	orange.400 #FB923C	#C2410C	#FB923C
Danger	red (z-top, overrides)	red.600 #DC2626	red.400 #F87171	#DC2626	#F87171

3.2 Design rationale per state

- **Disconnected** → **neutral grey, not red**. A clean, user-intended idle is **not** an alarm. Grey says "off, by your choice"; red is reserved for *danger*. This is exactly why the FFI splits Disconnected from Down [FFI §3.2] — the colour system honours that split (grey vs orange).
- **Connecting / Handshaking** → **amber**. Transient "working" states share one amber so the user reads a single "establishing..." phase, not two confusingly different colours.
- **Connected{direct}** → **green**, **Connected{relay}** → **green sub-shade (teal)**. Both are "protected = green-family", but a **direct** path (P2P, lower latency) shows pure green #15803D/#22C55E while a **relay** path (DERP-style) shows a teal-leaning green #0F766E/#2DD4BF. The user still reads "protected"; the StatusChip text ("direct" / "relay") and the sub-shade together convey the path without weakening the safety signal. Connected.path is supplied by the FFI projector [FFI §3.3].
- **Reconnecting** → **amber pulse**. Same amber as Connecting, animated via motion.semantic.connectPulse (1200 ms ease-in-out loop, [DT §6.4]). Under the OS reduce-motion flag it degrades to **static** amber — never strobes (ally + §11.4.107 no-flash).
- **Down** → **orange, distinct from Danger red**. An *unexpected* drop (ladder-exhausted / host-fatal [FFI §3.3]) is serious but not a *leak*; orange separates "it dropped, retrying/failed" from "you are exposed". Distinct hue, not just a shade, so colour-blind users still read two different alarms via the paired icon.
- **Danger** → **red, top of the z-stack**. Leak / kill-switch-tripped. Red **and** z-layer danger (2000, [DT §6.5]) so it can never be occluded; it overrides every other state colour, mirroring the FFI projector's "Danger overrides all intent" rule [FFI §3.3]. The kill-switch banner paints red.600/red.400 with a white label.

stateDiagram-v2

```

[*] --> Disconnected
Disconnected --> Connecting : start()
Connecting --> Handshaking : carrier up
Connecting --> Reconnecting : dial err (budget)
Handshaking --> Connected_direct : WG handshake (P2P)
Handshaking --> Connected_relay : WG handshake (relay)
Handshaking --> Danger : auth-failed (kill-switch closed)
Connected_direct --> Reconnecting : drop
Connected_relay --> Reconnecting : drop
Connected_direct --> Danger : leak / killswitch_tripped

```

Connected_direct --> Disconnected : user stop
 Reconnecting --> Connected_direct : re-dial ok
 Reconnecting --> Down : ladder-exhausted
 Down --> Connecting : start() again
 Danger --> Disconnected : user stop
 note right of Disconnected : grey #6B7385 / #4B5263
 note right of Connecting : amber #B45309 / #FBBF24
 note right of Connected_direct : green #15803D / #22C55E
 note right of Connected_relay : teal-green #0F766E / #2DD4BF
 note right of Down : orange #C2410C / #FB923C
 note right of Danger : red #DC2626 / #F87171 (z-top)

3.3 Two render patterns (and why both pass contrast)

Pattern	Where	Colour use	Governing WCAG threshold
Solid fill + label	ConnectButton (large circular), Danger banner	state colour as fill , text.onState (white, light theme) / #151920 (dark theme) as label	label is large+bold (≥ 24 px) → AA-large 3.0 governs, but every fill here clears AA-normal 4.5 too (§4.3)
Tint surface + coloured text/icon	StatusChip, list badges, inline status	state colour as text/icon on a low-alpha tint of itself (light) or on the dark surface (dark)	normal text → AA-normal 4.5 (§4.4)

The two patterns are why each state needs both a **dark stop** (light-theme fill

- light-theme text, low luminance reads on white) and a **bright stop** (dark-theme fill label is the dark #151920; dark-theme text reads on #151920). All proven in §4.

4. WCAG contrast proofs (computed)

WCAG 2.1 thresholds: **AA-normal 4.5**, **AA-large 3.0** (≥ 18 pt, or ≥ 14 pt bold), **AAA-normal 7.0**, **AAA-large 4.5**, **non-text UI/graphics 3.0** (SC 1.4.11). **Every ratio below is the real arithmetic** of the WCAG relative-luminance + $(L1+0.05)/(L2+0.05)$ contrast formula on the stated hexes (computed; not estimated — §11.4.6).

4.1 Text on surface — light theme

Pair	Ratio	AA-normal	AAA-normal
text.primary #151920 / surface.default #FFFFFF	17.62	✓	✓

Pair	Ratio	AA-normal	AAA-normal
text.primary #151920 / surface.raised #F7F8FA	16.58	✓	✓
text.secondary #4B5263 / #FFFFFF	7.82	✓	✓
text.tertiary #6B7385 / #FFFFFF	4.76	✓	× (AA only)
text.link #2F47C4 / #FFFFFF	7.45	✓	✓
text.disabled #9AA2B2 / #FFFFFF	2.57	exempt (inactive-control, SC 1.4.3)	—

4.2 Text on surface — dark theme (surface #151920)

Pair	Ratio	AA-normal	AAA-normal
text.primary #EDEFF3 / #151920	15.31	✓	✓
text.primary #EDEFF3 / surface.raised #222732	12.99	✓	✓
text.secondary #C2C8D4 / #151920	10.49	✓	✓
text.tertiary #9AA2B2 / #151920	6.87	✓	× (AA only)
text.link #9CB0FF / #151920	8.45	✓	✓
text.disabled #6B7385 / #151920	3.71	exempt (inactive-control)	—

4.3 Connection-state solid fills (label on fill) — both themes

Light theme: white #FFFFFF label. Dark theme: #151920 label. (Contrast is order-independent, so these are the proven white-on-fill / dark-on-fill ratios.)

State	Light fill	white-label ratio	AA-n	Dark fill	#151920-label ratio	AA-n
Disconnected	#6B7385	4.76	✓	#4B5263	— (uses brand idle, §3)	—
Connecting/ Handshaking/ Reconnecting	#B45309	5.02	✓	#FBBF24	10.56	✓
Connected{direct}	#15803D	5.02	✓	#22C55E	7.73	✓
Connected{relay}	#0F766E	5.47	✓	#2DD4BF	9.47	✓
Down	#C2410C	5.18	✓	#FB923C	7.79	✓
Danger	#DC2626	4.83	✓	#F87171	6.37	✓

All solid-fill state labels clear AA-normal (4.5) in both themes — the ConnectButton/Danger-banner label is legible even though it would only be held to AA-large (3.0) as large bold text. This is by design: the chosen light fills are the .600/.700 dark stops (not the raw .500), which is exactly why e.g. white-on-#16A34A (raw green .500, 3.30 — fails AA-normal) is **not** used as a light-theme fill; green.600 #15803D (5.02) is.

4.4 Connection-state text/icon on surface (tint pattern) — both themes

State	Light text/icon	on #FFFFFF	AA-n	Dark text/icon	on #151920	AA-n
Disconnected	#6B7385	4.76	✓	#9AA2B2	6.87	✓
Connecting	#B45309	5.02	✓	#FBBF24	10.56	✓
Connected{direct}	#15803D	5.02	✓	#22C55E	7.73	✓
Connected{relay}	#0F766E	5.47	✓	#2DD4BF	9.47	✓
Down	#C2410C	5.18	✓	#FB923C	7.79	✓
Danger	#DC2626	4.83	✓	#F87171	6.37	✓

Every connection-state text/icon clears **AA-normal (4.5)** in both themes; the dark-theme variants additionally clear AAA-normal (7.0) for all but Danger (6.37, AA).

4.5 Focus ring / non-text graphics (SC 1.4.11, floor 3.0)

Element	Light	Ratio vs surface	Dark	Ratio vs surface
border.focus ring	#3D5AF1 / #FFFFFF	5.34 ✓	#6B86FB / #151920	5.39 ✓

Captured-evidence note (§11.4.6). The 24 ratios cited in §4 were each produced by running the WCAG formula on the exact hex pair; the computation is reproducible and is the proof. No ratio is asserted without that arithmetic. The visual-regression suite ([visual-regression-and-qa.md], §11.4.162) re-runs these as automated contrast assertions so a future palette edit that drops a pair below its floor fails the build (gate CM-token-a11y-floor [DT §9]).

5. The no-overlap / no-label-overlay rule (§11.4.162)

§11.4.162 mandates "elements **MUST NOT** overlap or overlay labels". At the **colour layer** this is a concrete set of rules (layout enforcement is [component-library.md] + visual regression; this is the colour half):

1. **Never colour-on-colour that hides a label.** A label is **always** drawn on either a neutral surface (surface.*) or a *fill* whose contrast vs the label is proven (§4.3) — **never** a state colour text placed over another state colour fill. E.g. green "Connected" text is never painted on the amber "Connecting" chip during a

transition; the chip cross-fades fill **and** label together (`motion.stateXfade 180 ms`, [DT §6.4]) so no in-between frame shows an illegible colour-on-colour.

2. **Overlays use a scrim, never a translucent brand wash.** Any element that sits **over** content (modal, sheet, toast) places `overlay.scrim (#1F2937 @ 60 % light / #000000 @ 70 % dark)` beneath itself so its own labels meet contrast against an opaque-enough backdrop (white-on-scrim **14.68**, §4) — never a semi-transparent brand tint that lets busy content bleed through and reduce label contrast below the floor.
3. **State colour is reinforcement, layout owns separation.** Because the only saturated colour on screen is the active state (§0.2), two coloured elements are never adjacent enough to be confused; spacing tokens ([DT §6.1]) keep a `≥space.scale.2` (8 px) gap between any two coloured chips/badges so colour regions never touch (the colour-layer companion to the layout no-overlap rule).
4. **Danger is never overlaid by anything.** Z-layer danger (2000, [DT §6.5]) sits above modals/toasts; nothing can be drawn over a leak/kill-switch banner, so its label is never occluded.

These four rules are asserted by the visual-regression + token-drift suite (golden screenshots per theme, OCR/contrast checks, §11.4.162 / §11.4.168) in [visual-regression-and-qa.md]; this document supplies the colour contract they assert against.

6. Per-app accent differentiation (Client / Console / Connector)

All three apps share the **same** surface/text/feedback/connection-state semantics (safety colours never differ — D-COLOR-1). They differ **only** in the **brand-overridable** set `{action.primary, accent.*}` ([DT §5.3, D-DT-1]) so each app has a recognisable identity without ever recolouring a safety signal.

App	Accent ramp	Light accent (action/link)	Dark accent	Proof
Client (Helix Access)	<code>helix.*</code> indigo-blue (the brand)	<code>helix.600</code> #2F47C4	<code>helix.500</code> #3D5AF1 / <code>helix.400</code> #6B86FB	link 7.45 light; white-on-fill 7.45 ; dark accent 5.39
Console (admin)	<code>violet.*</code> (authority/ admin)	<code>violet.700</code> #6D28D9	<code>violet.400</code> #A78BFA	text 7.10 light; on <code>violet.600</code> #7C3AED white 5.70 ; dark 6.47
Connector (appliance)	<code>teal.*</code> (infrastructure/ network)	<code>teal.700</code> #0F766E	<code>teal.400</code> #2DD4BF	text 5.47 light; dark 9.47

All per-app accents clear AA-normal (4.5) for text in both themes. Verified ratios: Client link #2F47C4/white **7.45**; Console #6D28D9/white **7.10**; Connector #0F766E/white **5.47**; dark accents #6B86FB/#151920 **5.39**, #A78BFA/#151920 **6.47**, #2DD4BF/#151920 **9.47**.

D-COLOR-1 (restated). The accent override touches **only** `action.primary` + `accent.*.state.*` (connection), `feedback.*` (success/warning/error/info), and `text/surface/border` are **identical** across all three apps and any future tenant brand. A Console must never be able to recolour "Connected" away from green — that is a §11.4.66 operator decision, not a theme capability.

7. Swatch map

```
flowchart TB
    subgraph BRAND["Brand – helix.* (indigo-blue)"]
        b5["helix.500 #3D5AF1 (core)"]; b6["helix.600 #2F47C4 (action L)"]; b4["helix.400 #6B86FB (action D)"]
    end
    subgraph NEUT["Neutral 0..1000"]
        n0["#FFFFFF (surface L)"]; n9["#151920 (surface D / text L)"]; n1["#EDEFF3 (text D)"]; n5["#6B7385 (text.tertiary)"]
    end
    subgraph STATE["Connection state (7-variant ffi::TunnelStatus)"]
        st0["Disconnected · grey #6B7385 / #4B5263"]
        st1["Connecting/Handshaking/Reconnecting · amber #B45309 / #FBBF24"]
        st3["Connected·direct · green #15803D / #22C55E"]
        st3r["Connected·relay · teal-green #0F766E / #2DD4BF"]
        st5["Down · orange #C2410C / #FB923C"]
        st6["Danger · red #DC2626 / #F87171 (z-top)"]
    end
    subgraph FEED["Feedback"]
        f1["success #15803D / #34D399"]; f2["warning #B45309 / #FBBF24"]
        f3["error #DC2626 / #F87171"]; f4["info #1D4ED8 / #60A5FA"]
    end
    subgraph ACCENT["Per-app accent (brand-overridable only)"]
        a1["Client · helix #2F47C4 / #6B86FB"]
        a2["Console · violet #6D28D9 / #A78BFA"]
        a3["Connector · teal #0F766E / #2DD4BF"]
    end
    BRAND --> ACCENT
    STATE -. "amber pulse on Reconnecting; Danger overrides all" .-> STATE
```

Family	Light hex	Dark hex	Meaning
Brand	#2F47C4	#3D5AF1	trust, primary action
Disconnected	#6B7385	#9AA2B2	off (by choice)
Connecting/Reconnecting	#B45309	#FBBF24	working (amber, pulse)
Connected · direct	#15803D	#22C55E	protected, P2P
Connected · relay	#0F766E	#2DD4BF	protected, relayed

Family	Light hex	Dark hex	Meaning
Down	#C2410C	#FB923C	unexpected drop
Danger	#DC2626	#F87171	leak / kill-switch — exposed

8. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-COLOR-1	Safety semantics (<code>state.*</code> , <code>feedback.*</code>) are NOT brand/tenant-overrideable; only <code>action.primary+accent.*</code> differ per app. Widening = §11.4.66 operator decision.	recommended, surfaced
D-COLOR-2	Light-theme state fills use the .600/.700 dark stops (not raw .500) so every white-label fill clears AA-normal 4.5 (§4.3).	decided (proven)
D-COLOR-3	Reconnecting reuses the Connecting amber + a pulse; reduce-motion degrades pulse→static (never strobes).	decided
D-COLOR-4	Connected.path direct vs relay → pure-green vs teal-green sub-shade; both read "protected", path conveyed by sub-shade + StatusChip text.	decided
C-COLOR-A (consumes)	7-variant <code>ffi::TunnelStatus</code> vocabulary owned by [FFI §3.2]; §3 stays total over it (gate CM-token-state-coverage [DT §9]).	contract
C-COLOR-B (provides)	These hexes are the literal values [design-tokens.md]'s primitive tier references and [token-export-pipeline.md] emits.	contract
C-COLOR-C (provides)	The §4 contrast ratios are the assertions [visual-regression-and-qa.md] re-runs as automated ally gates (§11.4.162).	contract
U-COLOR-1 UNVERIFIED	The exact light-theme tint alpha for state-chip backgrounds (the % of the state colour mixed into the chip surface) is pinned by the component spec + golden-screenshot suite; stated patterns use the proven text-on-opaque-surface ratios, the more conservative bound.	open

Sources verified

- **Brand palette, neutral ramp, status/feedback/accent ramps, all hex choices, semantic light+dark mappings, connection-state palette, per-app accents** — **NO external source needed** — **original HelixVPN design work** (the hue choices and every hex are owned by this document).
- **WCAG contrast formula** ($(\text{relative luminance} + (L1 + 0.05)) / (L2 + 0.05)$) **and the AA/AAA/SC-1.4.11 thresholds** — *W3C Web Content Accessibility*

Guidelines 2.1, SC 1.4.3 (Contrast Minimum, incl. the inactive/disabled-control exception), SC 1.4.6 (Contrast Enhanced), SC 1.4.11 (Non-text Contrast), and the relative-luminance definition <https://www.w3.org/TR/WCAG21/#dfn-relative-luminance> (verified 2026-06-25). **Every ratio in §4, §6 was computed by this formula on the stated hexes** — the computation is the captured evidence (§11.4.6), not an estimate.

- **7-variant ffi::TunnelStatus vocabulary (Disconnected · Connecting · Handshaking · Connected{direct|relay} · Reconnecting · Down · Danger)** — final/v04-client/ffi-surface.md §3.2–§3.3 (read 2026-06-25).
- **Token tier/naming/theming model the colours live in** — final/v10-design/design-tokens.md (sibling, this wave).
- **3-app / 8-platform matrix, §11.4.162 design-system mandate (no-overlap / light+dark)** — final/SPECIFICATION.md §3 + MASTER_INDEX Volume 10 block (read 2026-06-25).
- The single UNVERIFIED item (U-COLOR-1, exact chip tint-alpha) is pending its named component/golden-screenshot spec per §11.4.6 — not asserted as fact.